RELATIONAL MODEL SCHEMA

Step 1 : Mapping of Regular Entity Types

ACCOUNT

AccountID	Email	PhoneNumber	Password	RegisterDate

STAFF

StaffID	StaffName	Role

MEMBER

MemberID	MemberName	Points

RESTAURANT_TABLE

TableID	Capacity	IsAvailable

MENU_ITEM

ItemID	ItemName	ItemType	ItemCategory	ItemPrice	ItemDescription

CARD_PAYMENT

CardID	AccountHolderName	CardNumber	ExpiryDate	SecurityCode

RESERVATION

ReservationID	CustomerName	ReservationDate	ReservationTime	HeadCount	SpecialRequest

KITCHEN_ORDER

KitchenID	Quantity	TimeSubmitted	TimeEnded

BILL

BillID	PaymentMethod	BillTime	PaymentTime

Step 2 : Mapping of Weak Entity Types

There is no weak entity type in the ER Diagram.

Step 3 : Mapping of Binary 1:1 Relationship Types

MEMBER

MemberID	MemberName	Points	*AccountID*

STAFF

StaffID	StaffName	Role	*AccountID*

BILL

BillID	PaymentMethod	BillTime	PaymentTime	*ReservationID*	*CardID*

Step 4 : Mapping of Binary 1:n and n:1 Relationship Types

RESERVATION

ReservationID	CustomerName	ReservationDate	ReservationTime	HeadCount	SpecialRequest	*TableID*

KITCHEN_ORDER

KitchenID	Quantity	TimeSubmitted	TimeEnded	*TableID*	*ItemID*

BILL

BillID	PaymentMethod	BillTime	PaymentTime	*StaffID*	*MemberID*	*ReservationID*	*TableID*	*CardID*

Step 5 : Mapping of Binary m:n Relationship Types

The relationship Contains between BILL and MENU_ITEM is m:n, so a new relation is created.

BILL_ITEM

BillItemID	*BillID*	*ItemID*	Quantity

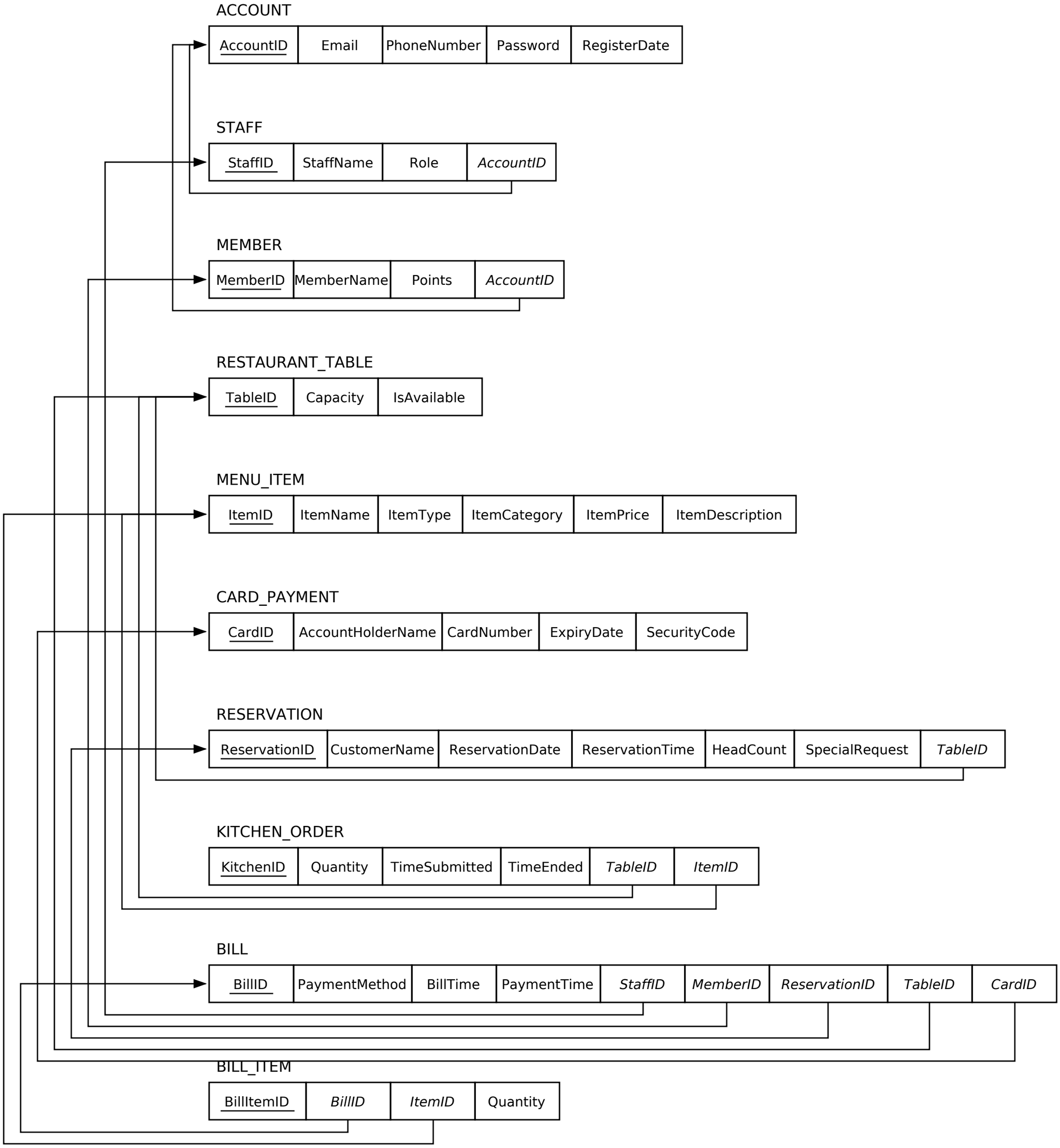
Step 6 : Mapping of Multi-valued Attributes

There is no multi-valued attribute in the ER Diagram.

Step 7 : Mapping of n-ary and Unary Relationship Types

There is no n-ary and unary relationship type in the ER Diagram.

FINAL RELATIONAL MODEL



Conclusion : From the above ER Diagram the relational schema is obtained as shown.